

Day 10: Advanced Data Modeling Techniques

[DEF] Definition to Remember

Advanced data modeling addresses complex relational patterns including Role-Playing Dimensions (multiple date keys), Inactive Relationships (activated via USERELATIONSHIP), and Bridge Tables for Many-to-Many resolution.

1. Role-Playing Dimensions & Inactive Relationships

- * Problem: Fact_Sales has both OrderDate, ShipDate, and DeliveryDate, all needing the Dim_Date table.

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2. Resolving Many-to-Many (*:*) Relationships via Bridge Tables

- * Instead of using native direct Many-to-Many relationships, introduce an intermediate distinct Bridge Table with 1:* relationships on both sides:

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3. Parent-Child Hierarchies (Organization Charts)

- * Flattening manager-employee hierarchies using DAX PATH() and PATHITEM() functions.

[MUST-WRITE] Must-Master Skills & Interview Points

1. When to use USERELATIONSHIP() with inactive relationships vs duplicating the Date dimension table.
2. How Bridge tables prevent Cartesian product inflation in Many-to-Many scenarios.
3. Best practice for handling multiple transaction dates in fact tables.

[RECALL] Quick Recall

Active Relationship (Default) + Inactive Relationship (Activated via USERELATIONSHIP) + Bridge Table (*:*)